

Article

Exploring the Application of Text-to-Image Generation Technology in Art Education at Vocational Senior High Schools in Taiwan

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Academic Editors: Joao Carlos Lopes Batista, Dimitris Apostolou and Anabela Mesquita

Received: 13 February 2025

Revised: 19 April 2025

Accepted: 22 April 2025

Published: 23 April 2025

Citation: Liao, C.-W.; Chen, H.-W.; Chen, B.-S.; Wang, I.-C.; Ho, W.-S.; Huang, W.-L. Exploring the Application of Text-to-Image Generation Technology in Art Education at Vocational Senior High Schools in Taiwan. *Information* **2025**, *16*, 341. <https://doi.org/10.3390/info16050341>

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Abstract: Exploring the potential of text-to-image generation technology in Taiwanese vocational high school art courses, this study employs a conceptual framework of technology integration, creative thinking, and metacognitive abilities, focusing on its effects on teaching strategies as well as students' digital art creation skills and cognitive and creative development. The study was conducted through a multi-methodological approach that includes a systematic literature review plus participatory action research and qualitative analysis. The results showed that integrating text-to-image technology with education boosted students' interest in activities such as prompt design and project creation and suited themes like landscapes and conceptual art. Testing AI tools enhanced technical proficiency (average of 3.95/5), while pedagogy shifted to project-based learning, increasing engagement. Students' digital art skills improved from 3.26 to 3.78 (16% growth), with creativity and originality (3.82/5), style diversity, visual complexity, and divergent thinking notably advanced. The technology also fostered metacognitive skills and critical thinking, proving to be an effective teaching aid beyond a mere digital tool. This discovery provides a fresh theoretical viewpoint and instructional procedures for high school art education curricula, anchored in technology, and highlights the importance of nurturing students' innovativeness and adaptability within the contemporary digital age.

Keywords: text-to-image generation technology; art education; digital art; creative thinking; critical thinking; vocational education

1. Introduction

Rapid advancements in artificial intelligence (AI) have brought transformative potential to a range of sectors, including education, where they are reshaping conventional paradigms through intelligent tutoring systems, adaptive learning tools, and data-driven pedagogical strategies. As AI technologies such as machine learning, computer vision, and natural language processing continue to evolve, they offer new pathways for enhancing creativity and personalizing learning experiences [1–4].

Within the field of visual arts education, text-to-image generation technologies—such as DALL-E, Midjourney, and Stable Diffusion—have recently emerged as powerful tools capable of translating textual prompts into visual outputs with remarkable speed and diversity [5–8]. This capability has prompted growing interest in how such tools might support, rather than supplant, the development of creative skills in educational settings. While existing research has examined AI's role in general education, less is known about its practical integration into vocational high school art programs, particularly in contexts where hands-on creativity and traditional skill development remain foundational.

This study investigates the integration of text-to-image generation technology into the art curriculum of vocational senior high schools in Taiwan. It specifically aims to achieve the following:

- (1) Discover the topics and types of activities that are suitable for integration with the technology.
- (2) Develop and test artificial intelligence models supporting the program of visual arts.
- (3) Study didactical changes which occur as a result of the implementation of this type of technology.
- (4) Analyze changes in students' abilities to create digital art after some time from starting the curriculum.

Evolving approaches in art education are increasingly incorporating digital tools to enhance both creative expression and technical proficiency, particularly in vocational settings where aligning with industry demands is essential [9]. As digitalization becomes more prevalent, striking a balance between traditional artistic practices and technological innovation has become a pressing issue for Taiwan's vocational senior high schools. Integrating such technologies into the curriculum not only helps students acquire practical skills, but also cultivates adaptability and interdisciplinary thinking [10]. This study is timely and necessary, offering insights into how text-to-image generation can enrich art education while supporting the creative growth of vocational students.

2. Literature Review

2.1. AI Text-to-Image Technology in Educational Settings

Text-to-image generation technology has advanced rapidly in recent years. Through large neural network models trained on image–text pairs, these tools can transform textual input into visual output, with applications spanning visual arts and design. The shift from earlier Generative Adversarial Networks (GANs) to modern Diffusion Models has significantly enhanced image quality and semantic accuracy [11].

In education, the integration of such tools has attracted growing attention. Educators have begun incorporating platforms like DALL-E, Midjourney, and Stable Diffusion into their teaching to help students translate descriptive language into visual artworks—enhancing composition skills and digital expression [5]. However, some scholars have raised concerns regarding authorship and originality in AI-generated works, arguing that despite their technical prowess, such creations may lack the conceptual depth and intentionality characteristic of human-made art [12].

2.2. Pedagogical Changes in Art Education Under AI Influence

The adoption of AI technologies is not only transforming artistic production but also reshaping pedagogical practices in art education. Traditional skill-focused instruction is gradually giving way to project-based learning (PBL), cross-media collaboration, and interactive exploration. Students are encouraged to engage in prompt design and hands-on experimentation, thus developing both digital literacy and artistic practice [13].

Moreover, digital technologies have become vital tools in boosting student motivation and engagement. With immediate visual feedback from AI-assisted creation, students can participate in iterative learning and self-revision, making learning more dynamic and reflective [14]. Overall, the integration of AI is shifting art education from a static, transmissive model to a more interactive, reflective, and collaborative process.

2.3. Fostering Creativity and Metacognition Through AI-Driven Art

Creativity is central to art learning and is commonly characterized by divergent thinking (generating multiple ideas) and convergent thinking (selecting the best ideas) [15]. According to Gardner (2011), the theory of multiple intelligences highlights how diverse cognitive strengths can enhance students' ability to generate and refine artistic concepts [16].

AI tools show particular promise in stimulating creative thinking and stylistic exploration. By adjusting prompts and comparing outputs, students can explore a variety of visual styles, deepening their sensitivity to artistic elements. Additionally, the use of AI tools encourages metacognitive development—students must monitor and adjust their creative strategies, thereby enhancing their capacity for self-regulation and reflective thinking [17].

2.4. Art Education in Taiwanese Vocational High Schools

Art education in Taiwan's vocational high schools emphasizes practical skills and alignment with industry needs, resembling the technical education models of Germany and Switzerland [18]. In response to the growing demand from the creative industries, there is an increasing push toward interdisciplinary integration and digital innovation in art curricula. This shift has been further accelerated by the rise in remote learning and the need to revitalize traditional course content [9].

This study focuses on vocational high schools in Taiwan, recognizing the dual necessity for students to develop creative competence and technological adaptability. The integration of text-to-image generation technology offers a timely and strategic response to this educational transformation, bridging traditional artistic skills with emerging digital tools.

3. Research Design

This research is a combination of mixed-methods approaches. It uses three components—a systematic literature review, participatory action research, and qualitative analysis. The reason for using this design is to obtain both a broad picture of what is already known and a deeper understanding of the lived experiences of individuals in the Taiwanese vocational high school context.

The study was performed in these three phases:

1. A systematic literature review and curriculum design were developed.
2. Implementation phase: action research and data collection.
3. Qualitative analysis and results integration.

This structure supports an iterative cycle of implementation and reflection—which helps the researchers to adjust the intervention based on continuous findings and feedback throughout the course of its implementation.

3.1. Data Collection

The research subject was examined comprehensively through the use of several data collection methods, as detailed below:

1. **Participatory Observation:** This involved classroom interactions, student reactions, and technology usage patterns. Structured observation sheets and field notes were used to record the observations.
2. **Semi-structured Interviews:** Students, teachers and school administrators were the participants. Technology usage experience, perceived learning outcomes, and reflections on the creative process are some of the themes that were covered during the interviews.
3. **Student Artwork Collection:** Pre-intervention, mid-term, and post-intervention are the points at which student artwork was collected. Traditional media artworks as well as AI-assisted creations were the types collected.

3.2. Research Participants

This study was conducted at a vocational senior high school located in Taichung City, central Taiwan. The school is recognized for its focus on technical and vocational education and is equipped with advanced facilities that support digital art and design instruction. A total of 30 students participated in the study. These students were selected through purposive sampling based on the following criteria:

- (1) Enrollment in art-related departments such as Fine Arts, Design, or Multimedia;
- (2) Spanning grades 10 through 12 to represent different stages of learning;
- (3) Possessing prior basic digital art experience but no formal exposure to AI-generated image tools.

The participants included 12 males and 18 females, as shown in Table 1. The balanced distribution across departments and grade levels ensured diverse perspectives in engagement with the experimental course. The detailed distribution of participants is shown in Table 1.

Table 1. Profile of student participants (n = 30).

Grade Level	Department	Number of Students	Gender (Male/Female)
10th Grade	Fine Arts	10	3/7
11th Grade	Design	10	4/6
12th Grade	Multimedia	10	5/5
Total	—	30	12/18

In addition to students, five teachers from the same school participated as evaluators. These teachers were from the Fine Arts, Design, and Multimedia departments and were selected based on their teaching experience (6–12 years) and prior involvement in curriculum design involving digital tools. The teachers did not use AI tools themselves during the course but served as independent evaluators using the assessment rubric developed in this study.

3.3. Experimental Course Design

The design of the course constitutes an innovative strategy to blend avant-garde AI technology into vocational senior high school art education. It involves merging theory with practice and critical self-assessment with a view to providing students with competences in the changing sphere of digital art production—as well as encouraging their interest in new technologies within the realm of art on all levels that affect creativity and appreciation.

The course is divided into four sections, with each concentrating on different details about creating pictures from text in art using AI. The first part is where the technology of AI art generation is introduced, giving an outline of the field and showing how images are created from descriptions. The second section deals more with the practical side, guiding students on how to come up with effective text prompts as they try out various AI platforms. During the third session, which involves project work, students are supposed to work independently to come up with images based on their preferred themes. In the last session there is a presentation of works created by students that involves peer assessment and a discussion on areas like the essence and limitations that AI tools have when it comes to the creation of art.

This design for the course is new and unique, because it integrates cutting-edge AI technology into art education for vocational senior high school students. The purpose of the course—which is an integration of theory with practice and critical reflection—is to prepare students for the changing world in digital art creation while also developing their ability to engage critically with new technologies in the art world.

The evaluation of student learning is based on three components: the final artwork produced (50%), classroom participation and contribution to discussions (30%), and a brief reflective report (20%). The multi-dimensional nature of this evaluation method ensures a complete assessment not only of technical skill but also understanding the concepts that underlie the work generated. For detailed criteria on assessing the final artwork, refer to Table 2 (assessment rubric for AI-assisted art creation).

Table 2. Assessment rubric for AI-assisted art creation.

Criteria	Excellent (5)	Good (4)	Satisfactory (3)	Needs Improvement (2)	Poor (1)
Creativity and Originality (25%)	Highly innovative; unique ideas	Creative; some novel ideas	Some creativity; few original ideas	Limited creativity; conventional	No original thinking; mimics
Technical Proficiency (25%)	Expert use; seamless integration	Proficient use; good integration	Competent use; basic integration	Basic use; little integration	Poor use; no integration
Conceptual Depth (25%)	Profound ideas; deep reflection	Good ideas; clear reflection	Adequate ideas; some reflection	Superficial ideas; minimal reflection	No clear concept; no reflection
Integration of AI Elements (25%)	Masterful blend; enhances vision	Effective blend; contributes	Adequate blend; some enhancement	Poor blend; unclear contribution	No blend; detracts

Source: data compiled in this research. Excellent (5): exceeds expectations in all aspects. Good (4): meets expectations with some exceptional elements. Satisfactory (3): meets basic expectations. Needs Improvement (2): falls short of expectations in several aspects. Poor (1): fails to meet basic expectations.

This curriculum setup is an innovation and it aims to bring together the latest AI technology with art education at the vocational high school senior level. The provision of both theoretical knowledge and practical application—in addition to critical reflection—seeks to prepare students for the changing realm of digital art production while also nurturing their ability to critically engage with new technologies in the world of art.

The development of this curriculum signifies an innovative stride in infusing high-end AI technology into art education among vocational senior high schools. The course emphasizes not only the cognitive aspects in theory, but also the psychomotor domain through practical application and the affective domain through critical reflection. It is designed for the purpose of helping students adapt to the rapid changes in digital art

production and, at the same time, developing their ability to critically interact with new technologies introduced in the field of art.

3.4. Total Score Calculation

- (1) Sum the scores for each criterion (each out of 5);
- (2) Multiply each criterion score by 5 (to convert to a percentage);
- (3) Add all percentages for a total score out of 100.

3.5. Interpretation of Total Score

- (1) 90–100: outstanding achievement.
- (2) 80–89: very good achievement.
- (3) 70–79: good achievement.
- (4) 60–69: satisfactory achievement.
- (5) Below 60: needs significant improvement.

3.6. Text-to-Image Generation Examples for Art Education

There were four lessons in vocational art education using text-to-image generation technology—illustrated by the following examples. Each lesson presents both favorable and unfavorable textual descriptions, with their image outputs depicting varied hues and shades of AI art creation technology. These instances not only show what the technology can do, but also stress (without saying it) that thinking out of the box is equally as important as being creative while using AI to produce artworks. Students are able to stepwise probe into the AI art generation process starting from ordinary applications, then move onto creative projects—where they have to come up with interesting ideas themselves—and finally, reach a stage where they need to question if using AI aligns well with their purpose of creating art. The samples provide clear, living examples for investigating how AI can be integrated into teaching art, keeping spirit of art alive throughout. The following tables (Tables 2–5) present detailed examples of text descriptions and their corresponding image results for each of the four lessons.

Table 3. Lesson 1: introduction to AI art generation technology.

Text Description	Image Result Description
Positive: ‘A serene landscape with a calm lake reflecting a sunset sky, surrounded by lush pine trees’	
Negative: ‘A chaotic cityscape at night with neon signs and crowded streets’	

Source: data compiled in this research.

3.7. Data Analysis

The data analysis in this study followed a mixed-methods approach that incorporated both quantitative and qualitative strategies. The primary goal was to evaluate changes in students’ performance and learning experiences throughout the experimental AI-assisted art course.

A total of 30 students’ artworks were evaluated at three key points: pre-intervention, mid-course, and post-course. The evaluation criteria included creativity and originality, technical proficiency, conceptual depth, and the integration of AI elements. Each artwork

was assessed using a standardized rubric (see Table 1), and the scores were analyzed to identify trends and improvements over time.

Table 4. Lesson 2: basic practice in text-to-image generation.

Text Description	Image Result Description
Positive: ‘A futuristic classroom with holographic displays and robot assistants, bright and colorful’	
Negative: ‘An old, abandoned classroom with broken furniture and peeling paint, dark and gloomy’	

Source: data compiled in this research.

Table 5. Lesson 3: creative project development.

Text Description	Image Result Description
Positive: ‘A whimsical garden where musical instruments grow like plants, under a rainbow sky’	
Negative: ‘A dystopian world where nature is replaced by mechanical replicas, in muted colors’	

Source: data compiled in this research.

To ensure consistency and reliability in the evaluation, five art teachers served as independent raters. They were trained to use the rubric and conducted scoring independently. Inter-rater agreement was established through calibration sessions and peer review.

Quantitative data were analyzed using descriptive statistics (means, standard deviations, score distributions), and time-series comparisons were conducted to measure changes in student performance across different stages of the course.

For interview data, thematic coding was conducted using NVivo software v15. Each participant’s response was categorized into major themes. The number of participants who expressed each theme was counted and reported descriptively (e.g., “24 out of 30 students”). These numbers were also sometimes expressed as percentages for readability (e.g., 80%), but they are not statistical estimates. Rather, they indicate the relative frequency of each theme across participants.

To avoid confusion, the revised Results section presents both the raw counts and illustrative quotes from participants, clarifying the qualitative nature of the findings.

Additionally, qualitative data were collected from student reflection reports, classroom observations, and semi-structured interviews. These data were coded thematically and triangulated with quantitative findings to offer a comprehensive interpretation of the students’ creative development, engagement with AI tools, and metacognitive growth.

4. Results

4.1. Creativity and Originality in Student Works

The performance of the students was good with respect to creativity and originality, garnering an average score of 3.82. An estimated 25% of the students scored more than 4.5 in this criterion, which reflects a high level of innovative capability—but around 15% scored less than 3, showing difficulties in coming up with unique ideas. Themes such as landscapes and conceptual art, along with activities like prompt design and project creation, were found to be suitable for technology integration.

An interesting pattern is that those students who scored highest in AI element integration tended also to have higher scores in creativity, hinting at the possibility that AI technology might have provoked their innovative thinking. Refer to Tables 6–9.

Table 6. Statistics with percentages and ranking.

Statistic Category	Value	Percentage	Rank
Participant Information			
Total Number of Students	30	-	-
Number of Evaluating Teachers	5	-	-
Average Scores by Criterion			
Technical Proficiency	3.95/5	79%	1
Creativity and Originality	3.82/5	76.4%	2
AI Element Integration	3.79/5	75.8%	3
Conceptual Depth	3.56/5	71.2%	4
Overall Statistics			
Overall Average Score	3.78/5	75.6%	-
Highest Average Score	4.65/5	93%	-
Lowest Average Score	2.90/5	58%	-
Standard Deviation	0.42	-	-

Table 7. Score distribution.

Score Range	Number of Students	Percentage
4.5–5.0	3	10%
4.0–4.49	8	26.7%
3.5–3.99	12	40%
3.0–3.49	5	16.7%
<3.0	2	6.6%

Table 8. Detailed statistics by criterion.

Evaluation Criteria	Mean	Median	Mode	Standard Deviation
Creativity and Originality	3.82	3.85	4.0	0.45
Technical Proficiency	3.95	4.0	4.0	0.38
Conceptual Depth	3.56	3.5	3.5	0.52
AI Element Integration	3.79	3.8	3.5	0.41

Table 9. Time series data.

Evaluation Criteria	Initial Average	Final Average	Improvement Percentage
Creativity and Originality	3.45	3.82	+10.7%
Technical Proficiency	3.30	3.95	+19.7%
Conceptual Depth	3.20	3.56	+11.3%
AI Element Integration	3.10	3.79	+22.3%
Overall	3.26	3.78	+16.0%

4.2. Technical Proficiency in AI-Assisted Art Creation

Reflecting Section 2.2’s emphasis on technology application in art education, testing existing AI tools significantly improved students’ technical proficiency (average of 3.95/5). More than 30% of them scored above 4.5, a rate that demonstrates their successful control over tools for generating images from text.

In general, the scores related to technical proficiency improved along with the development of the course—which is quite typical and indicates that with time, students became more acquainted with AI tools. But there about 10% of students could not manage the technical side, suggesting a need for extra help. For detailed statistics and progress over time, refer to Tables 6, 7, 10 and 11.

Table 10. Lesson 4: artwork presentation and reflection.

Text Description	Image Result Description
Positive: ‘A portrait that blends human and AI characteristics, showing harmony between technology and humanity’	
Negative: ‘A scene depicting the conflict between traditional art methods and AI-generated art, chaotic and divided’	

Source: data compiled in this research.

Table 11. Profile of teacher evaluators (n = 5).

Teacher ID	Department	Teaching Experience (Years)	Role in Research
T1	Fine Arts	8	Evaluator (Rater)
T2	Design	10	Evaluator (Rater)
T3	Multimedia	6	Evaluator (Rater)
T4	Fine Arts	12	Evaluator (Rater)
T5	Design	9	Evaluator (Rater)

4.3. Conceptual Depth and Reflective Thinking

Conceptual depth was the most challenging area in the evaluation, with an average score of 3.56. Echoing Section 2.3’s discussion of metacognitive abilities, while about 20% of students excelled in this aspect (scoring above 4.5), a considerable portion (about 25%) scored below 3.

This result suggests that students faced some difficulties in incorporating deep conceptual thinking into their artwork. This may reflect a need to strengthen the teaching of concept development and critical thinking in the curriculum. See Tables 6, 7, 10 and 11 for detailed statistics and progress data.

4.4. Integration of AI Elements in Artistic Vision

The integration of AI elements scored relatively well, with an average of 3.79. About 28% of students scored above 4.5 in this criterion, demonstrating their ability to effectively blend AI-generated content with their artistic vision. This is consistent with Section 2.4's discussion of vocational education challenges. However, approximately 18% of students scored below 3 in this aspect, indicating challenges in finding a balance between AI and traditional artistic techniques. Refer to Tables 6, 7, 10 and 11 for comprehensive data.

4.5. Overall Performance Analysis

Overall, students performed well, with an average score of 3.78. The score distribution showed a normal distribution, with most students (about 60%) scoring between 3.5 and 4.2.

Technical proficiency and creative originality were the two areas where students performed best, suggesting that the introduction of AI tools indeed stimulated students' creativity while also enhancing their technical skills. These findings directly support the first research objective by identifying both subject matter and instructional strategies that align effectively with the use of text-to-image tools in art education. The results in areas involving interaction with AI tools suggest that the tested models were effective in supporting students' digital art creation, thereby addressing the second research objective concerning the development and testing of AI tools for visual arts instruction. See Tables 10 and 11 for overall performance statistics.

4.6. Longitudinal Progress Analysis

By comparing student performance at the beginning and end of the course, we found that most students (about 75%) improved across all four criteria. On average, students' overall scores increased from 3.45 at the beginning of the course to 3.78 at the end, a 9.6% improvement.

The most significant areas of improvement were technical proficiency and AI element integration, reflecting students' increasing mastery of AI tools over time. Refer to Table 9 for detailed time series data.

These results demonstrate measurable progress in students' ability to create digital art over a defined instructional period, thereby directly addressing the fourth research objective concerning the identification of changes in digital art competencies following the implementation of AI-assisted learning.

4.7. Comparative Analysis with Traditional Art Courses

Although this study did not have a direct control group, by comparing our results with previous traditional art course grades in the school, we found the following:

- (1) Students in the AI-assisted course scored an average of 12% higher in creativity and originality.
- (2) The rate of improvement in technical proficiency was about 15% faster than in traditional courses.
- (3) Performance in conceptual depth was comparable to that in traditional courses, with no significant difference.

These comparative findings suggest that the integration of AI tools led to a meaningful shift in instructional outcomes and learning dynamics, indicating didactical changes in both

the pacing and focus of teaching—thus supporting the third research objective regarding the transformation of teaching practices as a result of AI technology implementation.

4.8. Evaluators' Perspectives on the Rubric and Student Performance

The five evaluating educators unanimously agreed that the rubric effectively captured key aspects of AI-assisted art creation. They particularly noted that the “AI Element Integration” criterion was crucial for evaluating student performance in this new art form.

Educators observed high levels of interest and engagement with AI tools among students, which may explain the high scores in creativity and technical proficiency. However, they also suggested strengthening guidance on concept development and critical thinking in future courses to improve student performance in conceptual depth.

5. Discussion

This study examined the integration of text-to-image generation technology in the art education curriculum of vocational senior high schools in Taiwan. The findings suggest that such tools can support students in developing their creativity, enhancing their digital art skills, and engaging more actively in project-based learning.

Students generally responded well to the inclusion of AI tools in assignments such as the creation of landscape, symbolic, and conceptual art. This observation is in line with earlier research showing that digital platforms can support ideation and visual experimentation [13]. The enthusiasm observed during prompt writing and theme exploration indicates that AI-assisted tools may contribute not only to visual production but also to the early, formative stages of the creative process.

The use of platforms such as Midjourney and DALL·E appeared to contribute to notable improvements in students' technical skills. These gains align with previous findings that technology-enriched environments can accelerate digital competency development [14]. The course structure, which emphasized iterative experimentation and prompt adjustment, may have played a role in reinforcing both technical accuracy and self-directed learning. In parallel, changes were noted in classroom dynamics. Teachers reported increased levels of student participation, independent inquiry, and collaboration. These developments suggest that the integration of AI shifted the pedagogical approach toward a more learner-centered model, where instructors take on more of a guiding role. Such a shift is consistent with emerging views on teaching that integrate technology into creative disciplines [11].

While technical improvements were evident, students' growth in metacognitive awareness and conceptual reasoning was less pronounced. Some participants demonstrated greater self-reflection and critical engagement with their creative choices, but others remained focused primarily on esthetic output. This imbalance reflects a broader challenge noted in the literature—helping learners move beyond the surface-level use of technology to consider its role in shaping artistic intent and identity [11].

These findings suggest that text-to-image generation technology holds meaningful potential for enhancing vocational art education, especially when combined with instructional strategies that support reflection and conceptual development. By encouraging students to engage more actively with both the creative process and the implications of using AI as part of their artistic practice, educators can help foster not only technical proficiency but also critical awareness. However, this study was limited by its relatively small sample size and the implementation at a single school. Future research could benefit from broader sampling across regions and disciplines, as well as longitudinal designs to better understand the long-term impact of AI integration on students' creative growth and readiness for the demands of the creative industries.

6. Conclusions

6.1. Integration of AI in Art Education

This investigation brought to light the fact that the use of AI text-to-image generation technology greatly improves creative and technical abilities among students in vocational high school art education in Taiwan. The activity, where traditional art blended with AI tools, was highly engaging, resulting in an average score of 3.82 for creativity and 3.95 for technical proficiency among students. This is clear evidence of the successful implementation of AI models, since there was a 19.7% increase noted specifically in technical proficiency during the study period; thus, it can be inferred from these results that integration with AI is especially effective when working on projects that would foster innovation as well as encourage technical exploration in art education.

6.2. Pedagogical Shifts and Student Skill Development

AI integration in art education drove pedagogical shifts toward project-based learning, art-technology fusion, and abstract thinking. Students scored an average of 3.79 in “AI Element Integration”, reflecting a strong ability to blend AI with their art, but their conceptual depth averaged 3.56, suggesting a need for enhanced critical thinking. Digital art skills rose from 3.26 to 3.78 (16% increase), with creativity increasing by 22.3% and originality by 10.7%. While AI boosts technical and creative growth, balancing it with traditional skills remains a key challenge, requiring focus on both.

6.3. Implications and Future Directions

This study suggests that AI integration into art education is likely to boost students’ innovative thinking as well as digital skills, as positive results were obtained in creativity and technical proficiency. But the lack of deep conceptual ideas, shown by low scores, highlights a need for more theoretical understanding plus critical reflection. The impact of integrating AI in the long term on students’ art development and readiness for careers in creative industries should be investigated by future research: this will help in determining where the focus lies between technical AI skills and traditional artistic techniques, as well as the need to deepen conceptual understanding. The work performed here lays a foundation for future development efforts around curricula that integrate AI with art education; it underscores the need to strike a balance between technological innovation and the principles of art that are unchanging over time—an area of research that has garnered much interest.

Author Contributions: C.-W.L., H.-W.C., B.-S.C., I.-C.W., W.-S.H. and W.-L.H. contributed meaningfully to this study. All authors have read and agreed to the published version of the manuscript.

Funding: The authors received no financial support for the research, authorship, and/or publication of this article.

Institutional Review Board Statement: This study was approved by the Department of Industrial Education and Technology at the National Changhua University of Education, Taiwan. The questionnaire data were categorized in accordance with the Ministry of Health and Welfare’s guidelines (5 July 2012, No. 1010265075). According to these guidelines, the study was exempt from requiring approval from an ethical review committee as it falls under the category of “research conducted in general educational environments for teaching evaluation or assessment of instructional techniques or effectiveness”.

Informed Consent Statement: Informed consent for participation was obtained from all subjects involved in this study.

Data Availability Statement: The data presented in this study are available on request from the corresponding author. The data are not publicly available due to privacy and confidentiality reasons.

Acknowledgments: This study acknowledges the technical support provided by the Department of Industrial Education and Technology at the National Changhua University of Education. The authors would like to express their sincere gratitude to Joao Carlos Lopes Batista, Anabela Mesquita, Dimitris Apostolou, Amanda Liu, Lily Yang, and the anonymous reviewers for their thorough review of our manuscript and for their numerous constructive comments and valuable suggestions.

Conflicts of Interest: The authors declare no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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